

Space age

The story of how OneWeb plans to build a space network of satellites around the Earth



Paypal blockade

Paypal over copyright concerns moves in to block VPN and SmartDNS user payments handicapping online payments. <http://digit.in/VPNDNS>

Akamai GIGA

If you're cursed by perennially buffering video feeds that seem to take an eternity and more to load, Akamai has a solution in the works that might be just what the doctor ordered. Managing a global network of over two lakh servers, Akamai operates one of the largest CDNs (content delivery networks) in the world right now. Companies such as Facebook, Adobe, Autodesk and several online entertainment portals use Akamai's server infrastructure to move extremely high volume of data across the world in extremely little time. Not only that, Akamai also offers services such as local caching of content especially for online video platforms which reduces buffering and video load times significantly. What's more, Akamai thinks that the Internet can function even more efficiently through a new technology they've developed and tested. It plans to give the Internet a severe dose of steroids, which can help unlock a whole new level of performance never seen before. Ever.

Akamai hasn't released a whole lot of technical nitty gritty of its GIGA protocol yet but according to tests it conducted across its global CDN servers, GIGA proved to boost Internet speed for most recipients by as much as 30 per cent. In fact, in regions like India, China and Bolivia, data moved faster by as much as 150 per cent thanks to GIGA, according to Akamai. This is a significant number which could help evolve the basic infrastructure of the Internet as it functions right now.

Just what exactly is GIGA? It's a new set of protocols that Akamai aims to replace TCP with completely. For those who don't know, TCP is the protocol on which fundamental Internet communications takes place currently, and it's been largely unchanged since it was first implemented back in 1974. That's a whopping 42 years ago! GIGA proposes several improvements over TCP which can help extract better efficiency standards when it comes to transferring data over the Internet. TCP is notoriously infamous for routinely labelling free or de-congested data routes between two servers or a client and server as full or

at capacity even when they're not. Also whenever the user's Internet connection isn't steady, TCP's prone to dropping links frequently. Both of these TCP inefficiencies are severely reduced in Akamai's GIGA, among other things.

What's more, Akamai has no intention of keeping GIGA all to itself as it has plans to openly release the revamped Internet protocol and hopes for broad adoption – both by end user devices and

to speak. While both UDP and TCP form the core of the Internet protocol suite as it was defined since the late 1970s, they differ at a fundamental level. Where TCP requires the client and server to establish a thorough handshake and data integrity is ensured down to the network layer, UDP does away with some of these overheads from the network layer (leaving it for applications to figure out these time-sensitive nitty gritty) thereby allowing



new server manufacturers. Akamai, in fact, aims to combine GIGA with QUIC, a Google developed software protocol which is geared to enhance Internet speeds as well.

Google QUIC

It doesn't take a genius to figure out why Google's taking the lead on trying to fix the Internet's fundamental architecture, seeing how the search engine giant has so much at stake in its online business. Ensuring the Internet improves isn't just to ensure Google's success but to safeguard the interests of the entire global online economy and to spur next generation of innovative technology. Which is what QUIC's all about.

Short for Quick UDP Internet Connections, QUIC relies on UDP (User Datagram Protocol) instead of TCP (Transmission Control Protocol) – the current backbone of Internet communications, so

for a better, speedier real-time connection compared to TCP.

However, QUIC aims to solve both the transport (network) layer as well as application layer problems encountered by sophisticated web applications. If you were accessing a website on your browser or watching a streaming video, currently before you land on the website you see the whole "Waiting / Establishing connection" routine in the bottom corner of your browser. The sequence of events may seem lengthy depending on your connection speed but it's also a factor of TCP's inefficiency – since it requires at least 2-3 handshakes between client (user) and server (destination) to establish a successful connection and starts serving requested data. Through QUIC, for new connections only 1 handshake is required, while repeat connection requests need no handshakes and these requests are